Aras **Valizadeh**

Computer Engineering Student · Machine Learning Researcher

Tehran, Iran

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Education

Shahid Beheshti University *Tehran, Iran*

B.Sc. in Computer Engineering *Sep. 2022 – Present*

- GPA: 19.13/20.00
- Ranked 1st among 90 students in the 2022-entry Computer Engineering cohort, based on cumulative GPA through the end of the eighth semester.
- Selected coursework: Advanced Programming (20/20), Introduction to Machine Learning (20/20), Operating Systems (20/20), Computer Architecture (20/20), Data Structures (20/20), Linear Algebra (20/20), Artificial Intelligence (19.75/20), Computer Networks (20/20), Probability and Statistics (20/20).

Allameh Helli 3 High School (Sampad) *Tehran, Iran*

Diploma in Mathematics *Sep. 2019 – Jun. 2022*

- Final average: 19.84/20
- Studied at a school affiliated with the National Organization for Development of Exceptional Talents (NODET).

Publications

Published

Salimi-Badr, A., Valizadeh, A., and Parchamijalal, M. M. 2026. "GRIFFIN: A Gated Relaxing Interaction-Aware Fuzzy Framework with Integrated Negation-Aware Neurocomputing to Learn Incomplete and Non-Separable Rules for Classification." *Neurocomputing*, 660: 131866. doi:10.1016/j.neucom.2025.131866

Valizadeh, A., Baroutian, B., Taraghi, F., Mohammadi, Z., Aghaei, A., and Ebrahimi Moghaddam, M. 2026. "An Extended Table Detection Framework for Enhanced Recognition of Persian Tables in OCR Systems." *2026 International Interdisciplinary Conference on Artificial Intelligence: Engineering, Health, Finance and Humanities (IICAI)*, Tehran, Iran, pp. 1–6. doi:10.1109/IICAI70155.2026.11620979

Under Review

Salimi-Badr, A., Valizadeh, A., Shakoo, A., and Foroutan, N. 2026. "Fuzzy Lightweight Learning Systems: Foundations, Taxonomy, and Future Directions." Manuscript under review at *Neurocomputing*. *Equal contribution and shared first authorship: A. Salimi-Badr and A. Valizadeh.*

Salimi-Badr, A., Hosseini, M., and Valizadeh, A. 2026. "K-FUSION: Kernelized Fuzzy Unstructured Systems with Interaction-aware Operators for Non-separable Rules." Manuscript under review at *Applied Soft Computing*.

Research Experience

Neuro-Fuzzy Machine Learning Research *Shahid Beheshti University*

Undergraduate Researcher *2025 – Present*

- Conduct research in the Neuro-Fuzzy Machine Learning Research lab, led by Dr. Salimi-Badr.
- Contributed to GRIFFIN's design, implementation, experimental evaluation, and manuscript preparation; published in *Neurocomputing*.
- Co-first author of *Fuzzy Lightweight Learning Systems: Foundations, Taxonomy, and Future Directions*, conducted in the lab and under review at *Neurocomputing*.
- Co-author of K-FUSION, conducted in the lab and under review at *Applied Soft Computing*.

Image Processing and Computer Vision Laboratory *Shahid Beheshti University*

Undergraduate Researcher *2026*

- Proposed the original research idea for an extended table detection framework aimed at improving the recognition of Persian tables in OCR systems.
- Designed and implemented the proposed framework and contributed to its experimental evaluation.
- Conducted the research at the Image Processing and Computer Vision Laboratory under the supervision of Dr. Mohammad Ebrahim Moghaddam.
- Served as first author on the resulting peer-reviewed paper, published in the 2026 International Interdisciplinary Conference on Artificial Intelligence (IICAI) and indexed in IEEE Xplore.

Teaching Experience

AI-Based Software Systems (SE4AI) — *Dr. Alidoost-Nia*
Head Teaching Assistant — *Fall 2026 – Present*

Principles of Robotics — *Dr. Salimi-Badr*
Teaching Assistant — *Fall 2026 – Present*

Introduction to Machine Learning — *Dr. Salimi-Badr*
Head Teaching Assistant — *Spring 2026*

Linear Algebra — *Dr. Malek*
Head Teaching Assistant — *Spring 2026*

Advanced Programming — *Dr. Vahidi-Asl*
Head Teaching Assistant — *Spring 2024 – Present*

Artificial Intelligence — *Dr. Salimi-Badr*
Teaching Assistant — *Fall 2025*

Artificial Intelligence — *Dr. Abdoos*
Head Teaching Assistant — *Spring 2025*

Digital Logic Circuits — *Dr. Mahdiani*
Teaching Assistant — *Fall 2024 – Spring 2025*

Discrete Mathematics — *Dr. Abdoos*
Head Teaching Assistant — *Fall 2024*

Computer Architecture — *Dr. Rahmati*
Teaching Assistant — *Spring 2024 – Fall 2024*

Data Structures — *Dr. Alidoost-Nia*
Teaching Assistant — *Spring 2024*

Guest Lectures & Workshops

Thread Programming. Guest Lecture, Advanced Programming Course.

Introduction to NumPy. Technical workshop, Linear Algebra Course.

Review of Artificial Intelligence Course Chapters. Five instructor-led review sessions, Artificial Intelligence Course.

End-to-End Machine Learning Project. Technical workshop, Introduction to Machine Learning Course.

Work Experience

Sama Soft — *Tehran, Iran*
Back-End Developer — *Aug. 2023 – Jan. 2024*

- Developed and tested microservice-based backend systems using Node.js and Express.js.
- Contributed to implementing and maintaining backend services and application APIs.

Selected Projects

Robust Vision-Based Sudoku Solver *GitHub*

- Developed an end-to-end pipeline using OpenCV and PyTorch to detect Sudoku grids, correct perspective, extract cells, and recognize printed digits with a CNN.
- Implemented constraint-aware search over alternative digit predictions, combining model probabilities with Sudoku constraints to recover valid, uniquely solvable boards.
- Projected solution digits onto the original image using an inverse perspective transformation, preserving the printed clues.

Scientific Paper Summarizer *GitHub*

- Built a PDF processing and semantic retrieval pipeline using text chunking, sentence embeddings, and ChromaDB to support scientific paper summarization and question answering.
- Integrated OpenAI and local Llama models into a Streamlit interface for document upload, summarization, and queries about methodology and results.

MIPS Pipeline Processor *GitHub*

- Designed and implemented a pipelined MIPS processor in Verilog, including the datapath and core components for pipelined instruction execution.
- Simulated instruction execution to check processor behavior across pipeline stages.

Academic Service & Leadership

Managing Editor - SBU CSE Publication March 2024 - June 2027
We are actively producing podcasts and journals for our faculty's publication, guided by Prof. Shams Aliee.

Secretary Professional Activities Committee – IEEE Iran Section Since 2025
Responsible for coordinating professional and technical activities within the IEEE Iran Section, including event organization, communication with members.

Head of Student Committee – Artificial Intelligence Conference (IAI) Served as a Head of the Student committee for the Shahid Beheshti University Artificial Intelligence Conference (IAI) under the supervision of Dr. Salimi-Badr.

Skills

Programming Languages: Python, Java, C, JavaScript, MATLAB, Assembly, Verilog, Arduino, SQL, Git

Frameworks/Libraries: PyTorch, OpenCV, NumPy, Pandas, Scikit-learn,Matplotlib, Node.js, Express.js, Flutter, Svelte

Soft Skills: Teamwork, Leadership, Problem Solving, Presentation

Languages

Languages **Persian (Native), Kurdish (Native), English (Upper-Intermediate, B2)**